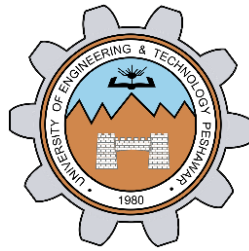


OPEN ENDED LAB PROJECT REPORT

Student Grade Book Management System

File Handling • Structs • GPA Calculation • Ranked Reports



SPRING 2026

CSE102L - COMPUTER PROGRAMMING LAB

Submitted by: **Muhammad Qasim Usman**

Registration No.: **25PWCSE2488**

Class Section: **C**

Submitted to:

Engr. Abdullah Hamid

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**Department of Computer Systems Engineering
University of Engineering and Technology, Peshawar**

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1. Abstract

The **Student Grade Book Management System** is a console-based application developed in C++ to simplify the management of student academic records. The system stores student information using structures and arrays, while file handling ensures permanent storage of records in a text file. It provides essential operations such as adding, searching, updating, and deleting student records with appropriate input validation.

The application automatically calculates subject grades and overall GPA according to a predefined grading scale, ranks students based on academic performance, and generates class reports and individual transcripts. In addition, it includes subject-wise statistical analysis, pass/fail determination, at-risk student identification, and personalized grade improvement suggestions to support academic evaluation. The project demonstrates the practical application of modular programming, file handling, structured data organization, sorting, searching, and decision-making techniques learned in the Computer Programming Lab.

2. Objective

The objective of this project is to develop a C++-based Student Grade Book Management System that efficiently manages student academic records using structures and file handling. The system automates GPA calculation, grade assignment, student ranking, report generation, and transcript creation while providing accurate record management through add, search, update, and delete operations. It reduces manual workload for instructors, minimizes calculation errors, and ensures organized, reliable, and permanent storage of student data.

3. File Format Design

The application stores student records in a text file named students.txt. Each line represents one student and contains the student's roll number, name, and marks of five subjects separated by spaces. The program reads each record sequentially, making the file simple to parse and update.

3.1. File Record Format

RollNo	Name	CP_Theory	Calculus	CA_Theory	CP_Lab	CA_Lab
101	Ali	85	78	88	78	85
102	Ahmad	43	56	87	0	9
103	Hamza	5	6	78	78	98

3.2. Sample Records

Field	Description
Roll No	Unique identification number of the student.
Name	Student name (single word or words joined using an underscore).
CP_Theory	Marks obtained in Computer Programming Theory.
Calculus	Marks obtained in Calculus.
CA_Theory	Marks obtained in Circuit Analysis Theory.

3.3. Format Explanation

Field	Description
CP_Lab	Marks obtained in Computer Programming Lab.
CA_Lab	Marks obtained in Circuit Analysis Lab.

3.4. Design Justification

A space-separated text format was selected because it is easy to read, edit, and process using C++ file handling. The sequential arrangement of fields allows the program to load records efficiently using stream extraction operators (>>). Although this format requires names containing spaces to be written with underscores, it simplifies parsing and reduces processing complexity.

4. Data Structure Design

The program uses two nested structures, Subject and Student, to organize academic information. The Student structure stores the student's personal and academic details, while each Subject structure stores information related to an individual subject. This design improves data organization and simplifies record management.

4.1 Structure: Subject

Field	Data Type	Approximate Size	Purpose
Name	String	Implementation-Dependent	Stores The Subject Name.
Marks	Float	4 Bytes	Stores The Marks Obtained In The Subject.
Grade	String	Implementation-Dependent	Stores The Letter Grade Calculated From The Marks.
GPA Points	Float	4 Bytes	Stores The GPA Points Corresponding To The Grade.

4.2 Structure: Student

Field	Data Type	Approximate Size	Purpose
Roll No	Int	4 Bytes	Stores the unique Roll No.of the student.
Name	String	Implementation-Dependent	Stores The Student's Name.
Subjects	Subject[5]	Depends On Subject Structure	Stores Academic Records Of Five Subjects.
GPA	Float	4 Bytes	Stores The Overall Gpa Calculated From All Subjects.

4.3 Design Justification

The nested structure design groups related information together, making the program easier to understand and maintain. Storing subject information inside each student record allows grades, GPA points, and marks to be managed efficiently. Additional fields such as grade, gpaPoints, gpa, and rank were included to eliminate repeated calculations and support features such as GPA computation, student ranking, report generation, transcript creation, and academic performance analysis.

5. Grading Scale

The system assigns a letter grade and corresponding GPA points according to the student's marks. The grading policy implemented in the program is based on a custom university-style grading scale, where each mark range is mapped to a specific letter grade and GPA value.

Marks Range	Grade	GPA Points
85 – 100	A	4.0
80 – 84	A–	3.7
75 – 79	B+	3.3
71 – 74	B	3.0
68 – 70	B–	2.7
64 – 67	C+	2.3
61 – 63	C	2.0
58 – 60	C–	1.7
54 – 57	D+	1.3
50 – 53	D	1.0
Below 50	F	0.0

5.1 Justification

This grading scale follows a university-style GPA system and is implemented directly in the program for automatic grade assignment and GPA calculation. The selected grade boundaries provide a consistent evaluation method and ensure that every student's academic performance is assessed uniformly throughout the system.

6. GPA Formula

The system calculates the overall Grade Point Average (GPA) by first assigning a letter grade and corresponding GPA points to each subject according to the predefined grading scale. After all subject GPA points are determined, the overall GPA is obtained by calculating their average. Since every student is evaluated in five subjects, each subject contributes equally to the final GPA.

6.1 GPA Formula

Overall GPA = (Sum of GPA Points of All Subjects) ÷ Number of Subjects

or

Overall GPA = (CP Theory + Calculus + CA Theory + CP Lab + CA Lab GPA Points) ÷ 5

6.2 Manual Verification

To verify the correctness of the GPA calculation, the following student record was selected from the `students.txt` file.

Subject	Marks	Grade	GPA Points
CP Theory	100	A	4.0
Calculus	99	A	4.0
CA Theory	98	A	4.0
CP Lab	97	A	4.0
CA Lab	96	A	4.0

6.3 Calculation:

Overall GPA = $(4.0 + 4.0 + 4.0 + 4.0 + 4.0) \div 5$

Overall GPA = $20.0 \div 5$

Overall GPA = 4.0

The manually calculated GPA is **4.0**, which is identical to the GPA generated by the program. This confirms that the GPA calculation method implemented in the system is accurate and reliable.

7. Program Flow Diagram

Figure 1 illustrates the overall execution flow of the Student Grade Book Management System. When the program starts, it automatically loads student records from the **students.txt** file into memory. After successful loading, the main menu is displayed, allowing the user to perform different operations such as displaying student records, searching, adding, updating, deleting, generating reports, generating transcripts, and viewing subject-wise statistics.

Operations that modify student data automatically update the **students.txt** file to ensure permanent storage. After completing any operation, the program returns to the main menu and continues execution until the user selects the **Exit** option, which terminates the application.

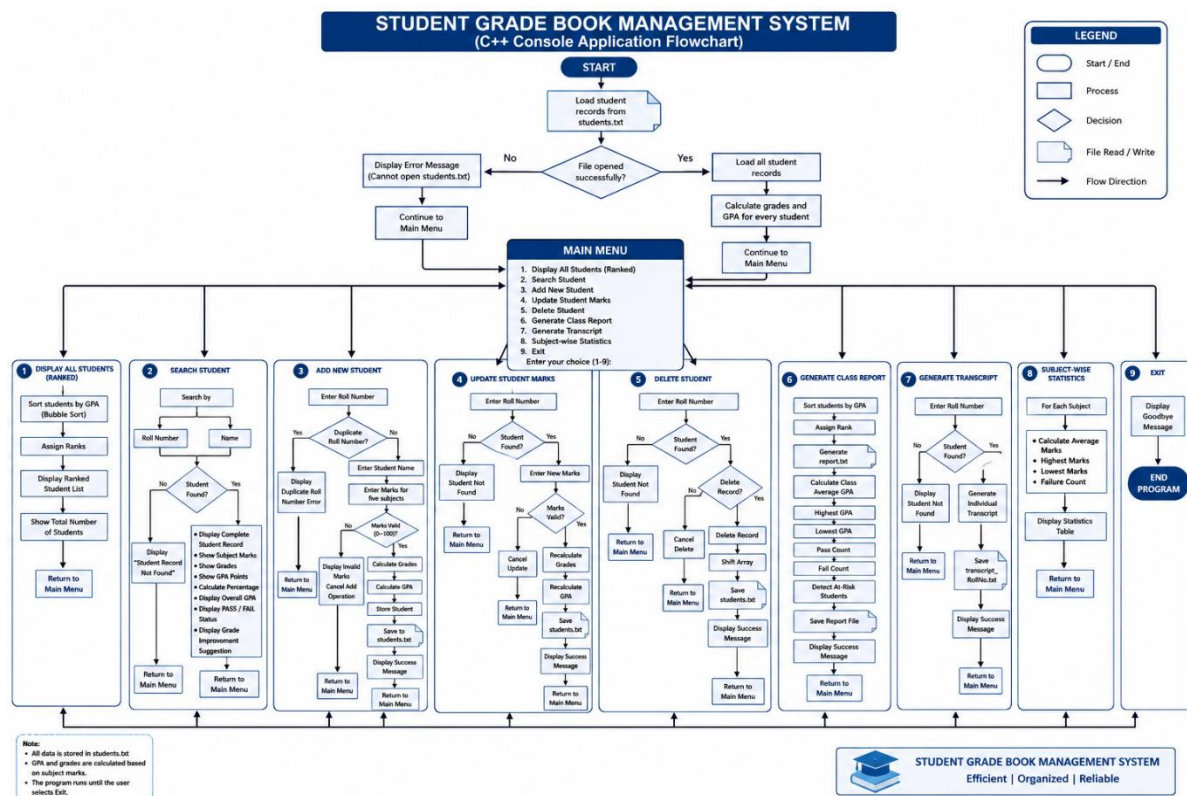


Figure 1: Overall Program Flow of the Student Grade Book Management System

8. Function Descriptions

The program follows a modular programming approach in which each function performs a specific task. This improves code readability, simplifies debugging, and makes the system easier to maintain and extend.

Function Name	Parameters	Return Type	Description
calculateSubjectGrade()	Subject &sub	void	Calculates the letter grade and GPA points for a subject based on the obtained marks.
computeStudentGPA()	Student &st	void	Computes the overall GPA by averaging the GPA points of all five subjects.
loadStudents()	None	void	Loads student records from students.txt , initializes subject names, and computes grades and GPA for each student.
saveStudents()	None	void	Saves all student records stored in memory back to students.txt .
sortByGPA()	None	void	Sorts students in descending order of GPA using Bubble Sort and assigns class ranks.
displayAll()	None	void	Displays the ranked list of students with marks, GPA, and rank information.
searchStudent()	None	void	Searches for a student by roll number or name and displays complete academic details along with percentage, pass/fail status, and improvement suggestions.
addStudent()	None	void	Adds a new student after validating input data, calculates GPA, and stores the record permanently.
updateMarks()	None	void	Updates the marks of an existing student, recalculates grades and GPA, and saves the updated record.

deleteStudent()	None	void	Deletes a student record after user confirmation and updates the data file.
generateReport()	None	void	Generates report.txt containing ranked students, GPA statistics, pass/fail summary, and at-risk student information.
generateTranscript()	None	void	Creates an individual transcript file containing marks, grades, GPA, percentage, rank, and final status.
subjectStatistics()	None	void	Computes subject-wise average, highest marks, lowest marks, and failure count for all students.
improvementSuggestion()	Student st	void	Identifies the weakest subject and recommends the marks required to achieve the next grade.
main()	None	int	Controls the overall execution of the program by loading records, displaying the menu, processing user selections, and terminating the application.

9. Challenges & Solutions

During the development of the Student Grade Book Management System, several technical challenges were encountered. These challenges required careful implementation to ensure the system remained reliable, accurate, and easy to maintain.

Challenge	Solution Implemented
Maintaining unique student records	Before adding a new student, the program checks all existing roll numbers. If a duplicate is found, the insertion is cancelled and an error message is displayed, ensuring every student has a unique identity.
Validating user input	The system verifies that marks entered for every subject are within the valid range (0–100). Invalid input immediately cancels the operation, preventing incorrect academic records.
Accurate GPA and grade calculation	Separate functions were developed to calculate subject grades and overall GPA automatically. This modular approach eliminates repeated calculations and ensures consistent results throughout the program.
Ranking students correctly	Bubble Sort was implemented to arrange students in descending order of GPA. After sorting, ranks are assigned automatically according to their academic performance.
Permanent data storage	File handling was used to load records from students.txt at program startup and save all modifications after add, update, or delete operations, ensuring data persistence between program executions.
Generating academic reports	Dedicated functions were developed to generate class reports and individual transcripts automatically, reducing manual effort and presenting academic information in an organized format.

10. Extensions Implemented

The basic Grade Book System was enhanced by implementing several additional features that improve data management, academic analysis, and reporting capabilities. These extensions make the application more practical and user-friendly.

Extension	Implementation
Extension A Delete Student Record	The system allows an instructor to delete a student record using the roll number. Before deletion, the program requests confirmation to prevent accidental removal. After successful deletion, the remaining records are shifted, the updated data is saved to students.txt , and student rankings are recalculated whenever the ranked list is displayed.
Extension C Transcript Generator	The program generates an individual transcript for a selected student. Each transcript includes the student's roll number, name, rank, percentage, overall GPA, subject-wise marks, grades, and pass/fail status. The transcript is automatically saved as transcript_RollNo.txt .
Extension D Subject-wise Statistics	The system analyzes the academic performance of each subject by calculating the class average, highest marks, lowest marks, and the total number of failed students. This feature provides instructors with a comprehensive overview of subject performance and helps identify subjects where students require additional support.
Extension E Grade Improvement Suggestions	After displaying a student's academic record, the program identifies the weakest subject and calculates the minimum marks required to achieve the next higher grade. If the student has already obtained the highest grade, the system displays an appreciation message instead of a recommendation.

The implemented extensions significantly improve the functionality of the Student Grade Book Management System. They provide permanent record management, automated transcript generation, detailed subject performance analysis, and personalized academic guidance, making the system more efficient and beneficial for both instructors and students.

11. Conclusion

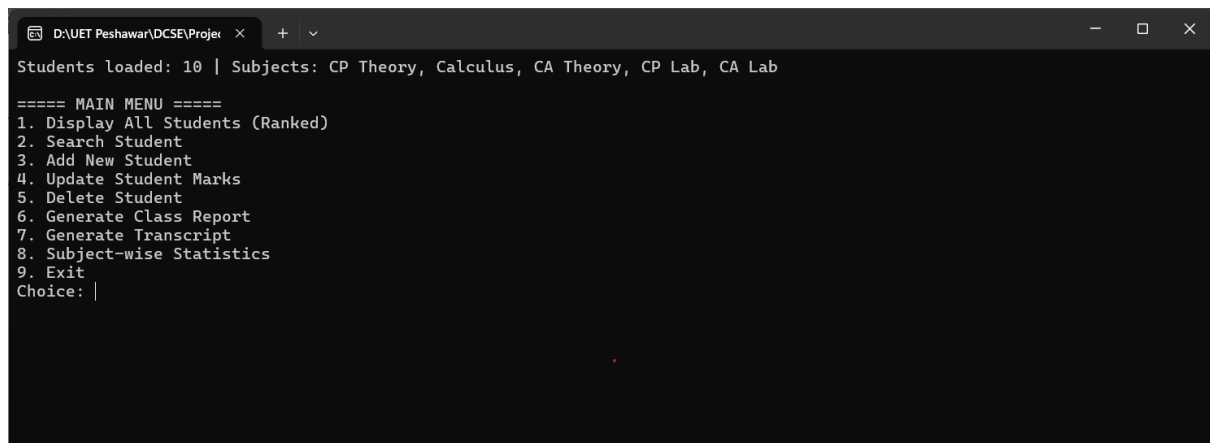
The development of the Student Grade Book Management System provided practical experience beyond the concepts covered in lectures. It strengthened my understanding of file handling, nested structures, modular programming, searching, sorting, input validation, and integrating multiple programming concepts into a complete application. The project also demonstrated the importance of organizing code into reusable functions and maintaining accurate data through proper validation and file management.

If the system were to be improved in the future, features such as a graphical user interface (GUI), support for multiple semesters, advanced searching and filtering, secure user authentication, and a database management system could be added. These enhancements would increase the scalability, usability, and overall functionality of the application.

11. PROGRAM OUTPUTS

11.1 Program Startup

- Program execution after loading **students.txt**.
- Main menu displayed successfully.



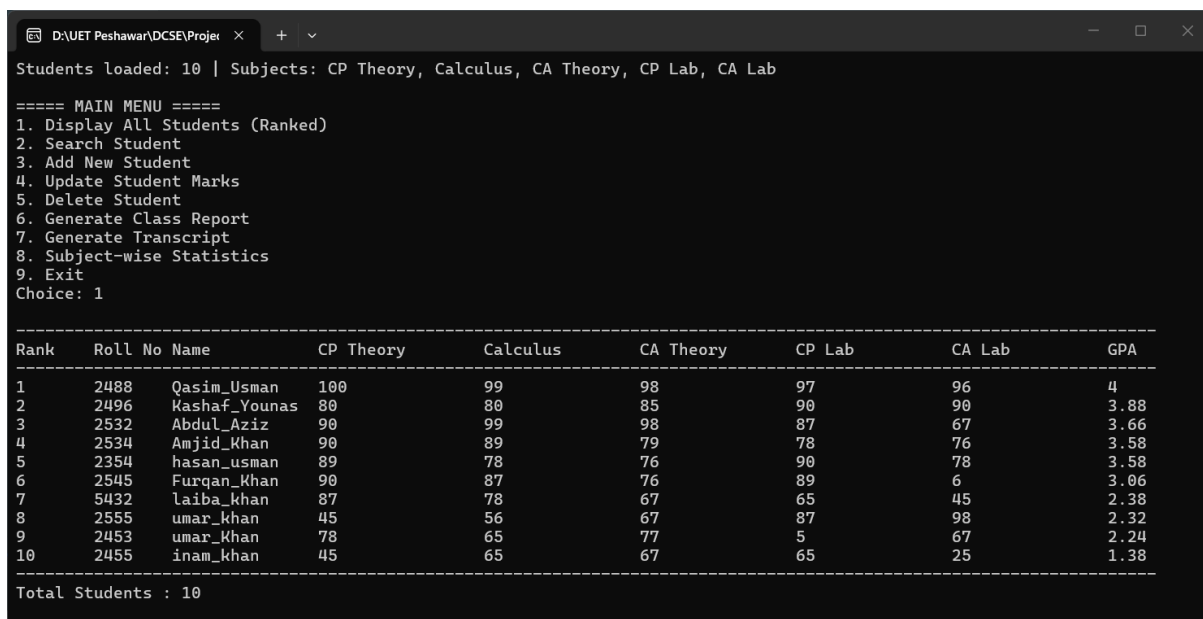
```
D:\UET Peshawar\DCSE\Projer x + v
Students Loaded: 10 | Subjects: CP Theory, Calculus, CA Theory, CP Lab, CA Lab

===== MAIN MENU =====
1. Display All Students (Ranked)
2. Search Student
3. Add New Student
4. Update Student Marks
5. Delete Student
6. Generate Class Report
7. Generate Transcript
8. Subject-wise Statistics
9. Exit
Choice: |
```

Figure 11.1: Program Startup and Main Menu

11.2 Display All Students

- Ranked list of all students after GPA calculation.
- Bubble Sort ranking and rank assignment.



```
D:\UET Peshawar\DCSE\Projer x + v
Students Loaded: 10 | Subjects: CP Theory, Calculus, CA Theory, CP Lab, CA Lab

===== MAIN MENU =====
1. Display All Students (Ranked)
2. Search Student
3. Add New Student
4. Update Student Marks
5. Delete Student
6. Generate Class Report
7. Generate Transcript
8. Subject-wise Statistics
9. Exit
Choice: 1

-----
Rank  Roll No  Name      CP Theory  Calculus  CA Theory  CP Lab  CA Lab  GPA
-----
1      2488    Qasim_Usman  100        99        98        97        96        4
2      2496    Kashaf_Younas  80        80        85        90        90        3.88
3      2532    Abdul_Aziz    90        99        98        87        67        3.66
4      2534    Amjid_Khan    90        89        79        78        76        3.58
5      2354    hasan_usman   89        78        76        90        78        3.58
6      2545    Furqan_Khan   90        87        76        89        6        3.06
7      5432    laiba_khan    87        78        67        65        45        2.38
8      2555    umar_khan     45        56        67        87        98        2.32
9      2453    umar_khan     78        65        77        5        67        2.24
10     2455    inam_khan     45        65        67        65        25        1.38
-----
Total Students : 10
```

Figure 11.2: Ranked List of Students

11.3. Search Student

11.3.1. Search Student by Roll Number

- Search using a valid roll number.
- Displays complete academic record, grades, GPA, percentage, pass/fail status, and improvement suggestion.

```
D:\UET Peshawar\DCSE\Proje... x + v

===== MAIN MENU =====
1. Display All Students (Ranked)
2. Search Student
3. Add New Student
4. Update Student Marks
5. Delete Student
6. Generate Class Report
7. Generate Transcript
8. Subject-wise Statistics
9. Exit
Choice: 2
Search by: (1) Roll Number (2) Name: 1
Enter roll number: 2488

--- STUDENT RECORD FOUND ---
Roll No : 2488
Name : Qasim_Usman
Rank : 1
Subject Marks    Grade    GPA Pts
CP theory        100     A      4
Calculus         99      A      4
CA theory        98      A      4
CP lab 97        A      4
CA lab 96        A      4
Overall GPA : 4
Percentage : 98%
Status : PASS

-----
GRADE IMPROVEMENT SUGGESTION
-----
Weakest Subject : CA lab
Current Marks : 96
Current Grade : A
Excellent! You already have the highest grade.
```

Figure 12.3: Search Student by Roll Number

11.3.2. Search Student by Name

- Search using the student's name.
- Displays the complete academic record.

```
D:\UET Peshawar\DCSE\Proje... x + v

===== MAIN MENU =====
1. Display All Students (Ranked)
2. Search Student
3. Add New Student
4. Update Student Marks
5. Delete Student
6. Generate Class Report
7. Generate Transcript
8. Subject-wise Statistics
9. Exit
Choice: 2
Search by: (1) Roll Number (2) Name: 2
Enter Name: Qasim_Usman

--- STUDENT RECORD FOUND ---
Roll No : 2488
Name : Qasim_Usman
Rank : 1

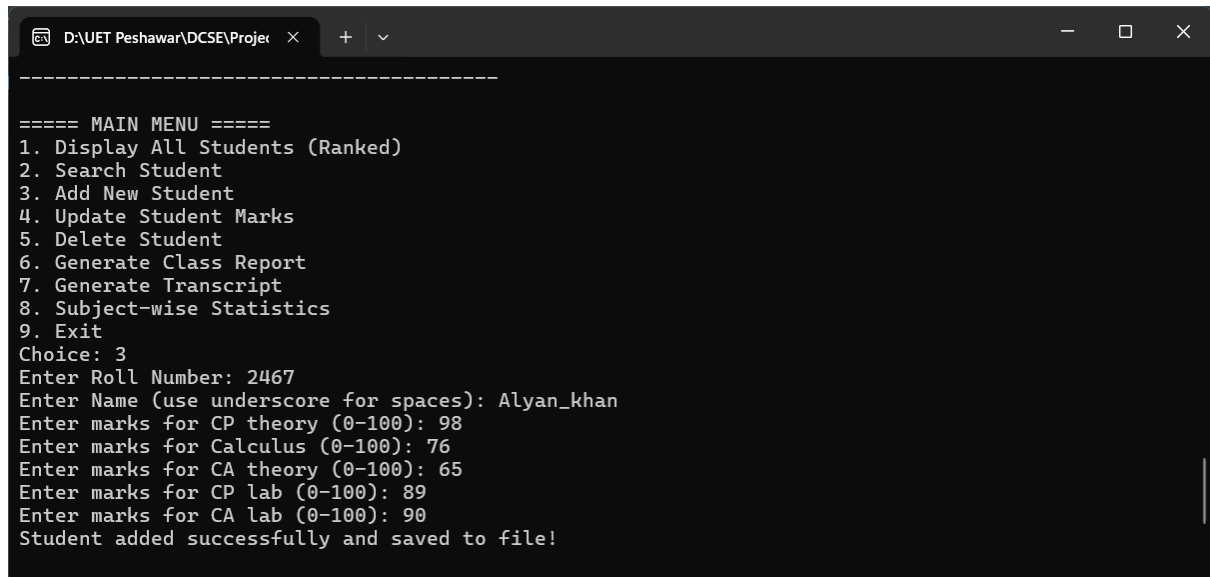
Subject    Marks    Grade    GPA Points
CP theory  100     A      4
Calculus   99      A      4
CA theory  98      A      4
CP lab 97  A      4
CA lab 96  A      4
Overall GPA : 4
Percentage : 98%
Status : PASS

-----
GRADE IMPROVEMENT SUGGESTION
-----
Weakest Subject : CA lab
Current Marks : 96
Current Grade : A
Excellent! You already have the highest grade.
```

Figure 11.3.3: Search Student by Nam

11.4. Add New Student

- Successful addition of a new student.
- Record stored permanently in students.txt.

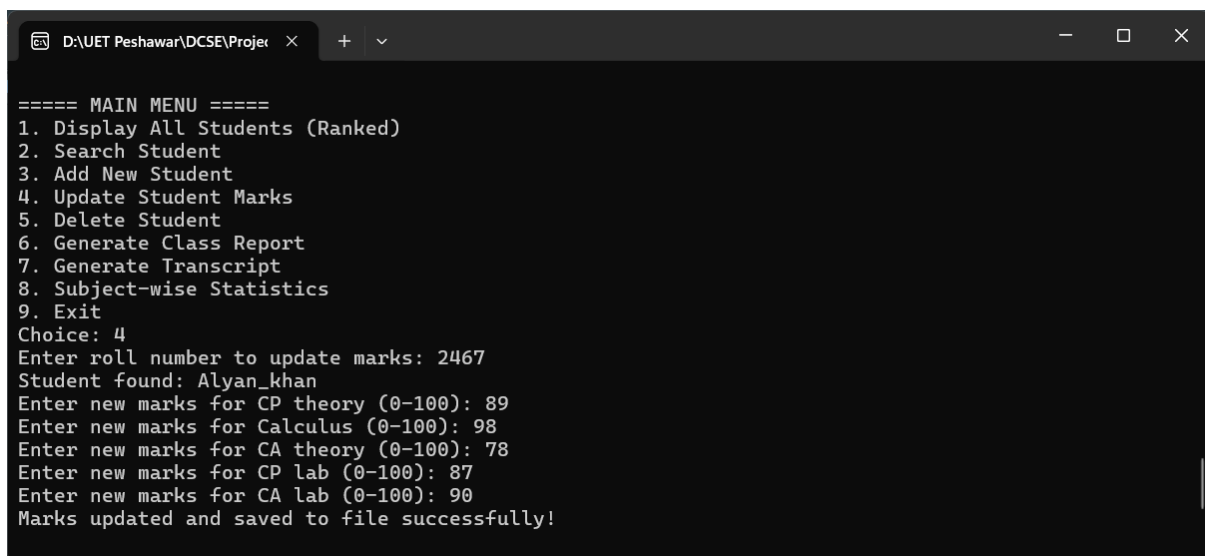


```
=====
===== MAIN MENU =====
1. Display All Students (Ranked)
2. Search Student
3. Add New Student
4. Update Student Marks
5. Delete Student
6. Generate Class Report
7. Generate Transcript
8. Subject-wise Statistics
9. Exit
Choice: 3
Enter Roll Number: 2467
Enter Name (use underscore for spaces): Alyan_khan
Enter marks for CP theory (0-100): 98
Enter marks for Calculus (0-100): 76
Enter marks for CA theory (0-100): 65
Enter marks for CP lab (0-100): 89
Enter marks for CA lab (0-100): 90
Student added successfully and saved to file!
```

Figure 11.4: Add New Student

11.5. Update Student Marks

- Existing student marks updated.
- GPA recalculated automatically.

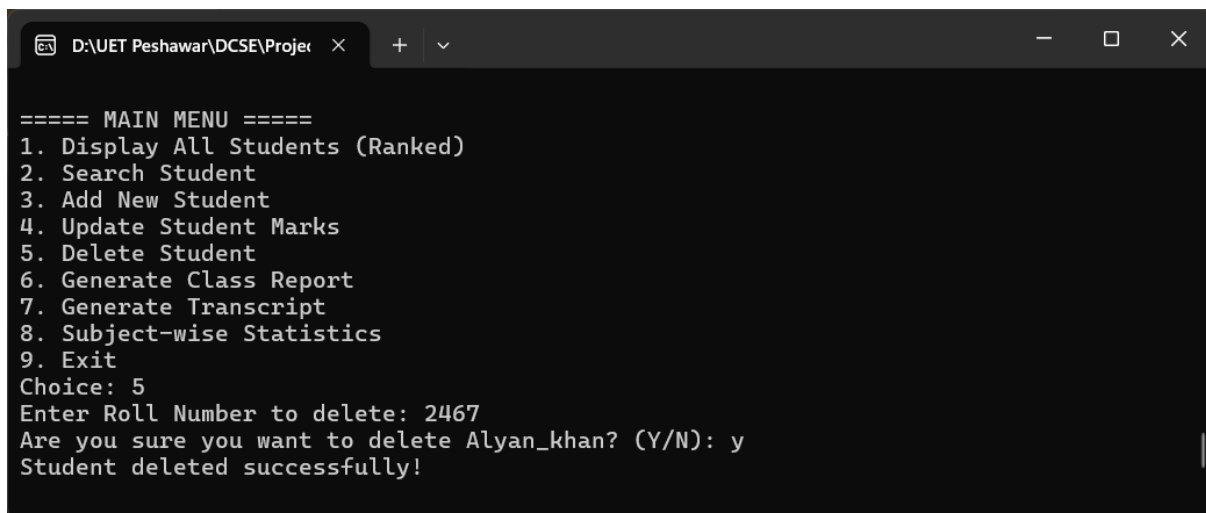


```
===== MAIN MENU =====
1. Display All Students (Ranked)
2. Search Student
3. Add New Student
4. Update Student Marks
5. Delete Student
6. Generate Class Report
7. Generate Transcript
8. Subject-wise Statistics
9. Exit
Choice: 4
Enter roll number to update marks: 2467
Student found: Alyan_khan
Enter new marks for CP theory (0-100): 89
Enter new marks for Calculus (0-100): 98
Enter new marks for CA theory (0-100): 78
Enter new marks for CP lab (0-100): 87
Enter new marks for CA lab (0-100): 90
Marks updated and saved to file successfully!
```

Figure 12.6: Update Student Marks

11.6. Delete Student Record

- Student deleted after confirmation.
- Record removed permanently from students.txt.

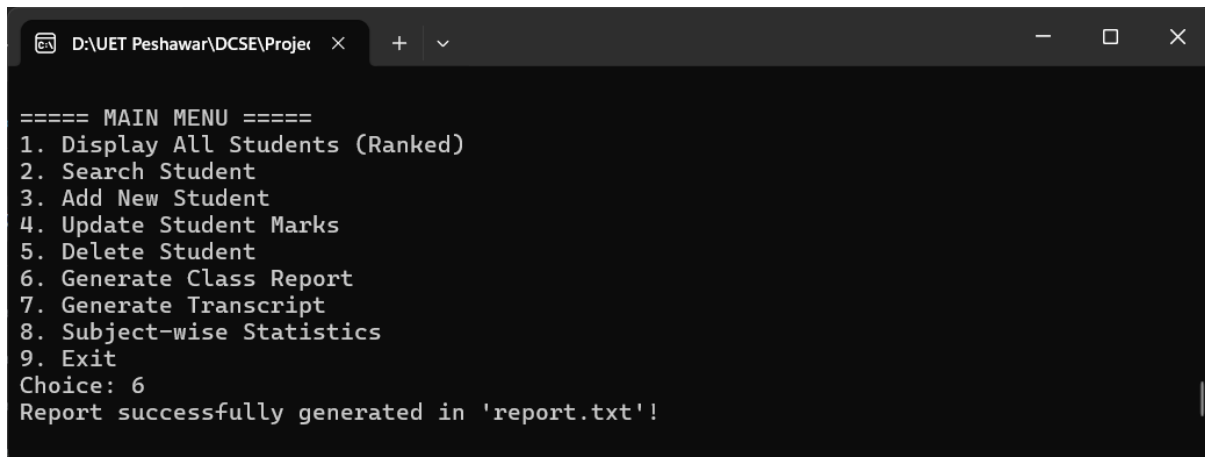
A screenshot of a terminal window with a dark background. The window title bar shows a file path: "D:\UET Peshawar\DCSE\Proje...". The terminal content displays a menu with 9 options. Option 5, "Delete Student", is selected. The user enters the roll number "2467". A confirmation prompt asks "Are you sure you want to delete Alyan_khan? (Y/N):" and the user responds with "y". The final output is "Student deleted successfully!".

```
==== MAIN MENU ====
1. Display All Students (Ranked)
2. Search Student
3. Add New Student
4. Update Student Marks
5. Delete Student
6. Generate Class Report
7. Generate Transcript
8. Subject-wise Statistics
9. Exit
Choice: 5
Enter Roll Number to delete: 2467
Are you sure you want to delete Alyan_khan? (Y/N): y
Student deleted successfully!
```

Figure 11.6: Delete Student Record

11.7 Generate Class Report

Successful generation of report.txt.

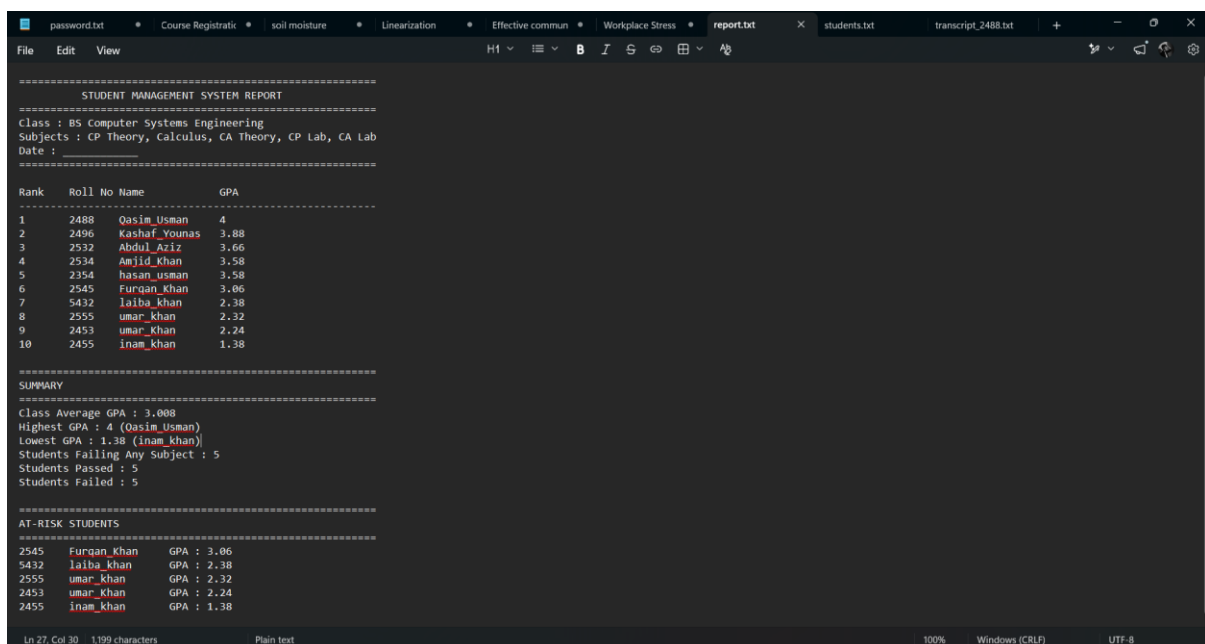


```
===== MAIN MENU =====
1. Display All Students (Ranked)
2. Search Student
3. Add New Student
4. Update Student Marks
5. Delete Student
6. Generate Class Report
7. Generate Transcript
8. Subject-wise Statistics
9. Exit
Choice: 6
Report successfully generated in 'report.txt'!
```

Figure 11.7: Console Output for Report Generation

11.8 Generated Class Report File

Contents of the generated report.txt file.



```
=====
STUDENT MANAGEMENT SYSTEM REPORT
=====
Class : BS Computer Systems Engineering
Subjects : CP Theory, Calculus, CA Theory, CP Lab, CA Lab
Date : _____
=====

Rank  Roll No Name      GPA
-----
1      2488  Qasim Usman      4
2      2496  Kashaf Younas    3.88
3      2532  Abdul Aziz       3.66
4      2534  Amjid Khan       3.58
5      2354  hasan_usman      3.58
6      2545  Furqan Khan      3.06
7      5432  Jaiba Khan       2.38
8      2555  umar_khan        2.32
9      2453  umar_khan        2.24
10     2455  Inam Khan        1.38
=====

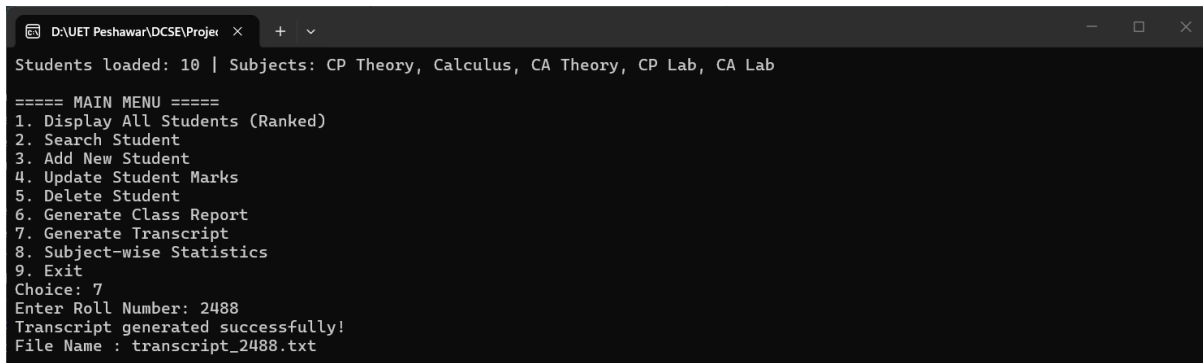
SUMMARY
=====
Class Average GPA : 3.008
Highest GPA : 4 (Qasim Usman)
Lowest GPA : 1.38 (Inam Khan)
Students Failing Any Subject : 5
Students Passed : 5
Students Failed : 5

=====
AT-RISK STUDENTS
=====
2545  Furqan Khan      GPA : 3.06
5432  Jaiba Khan       GPA : 2.38
2555  umar_khan        GPA : 2.32
2453  umar_khan        GPA : 2.24
2455  Inam Khan        GPA : 1.38
=====
```

Figure 11.8: Generated Class Report

11.9 Generate Student Transcript

Successful transcript generation message.



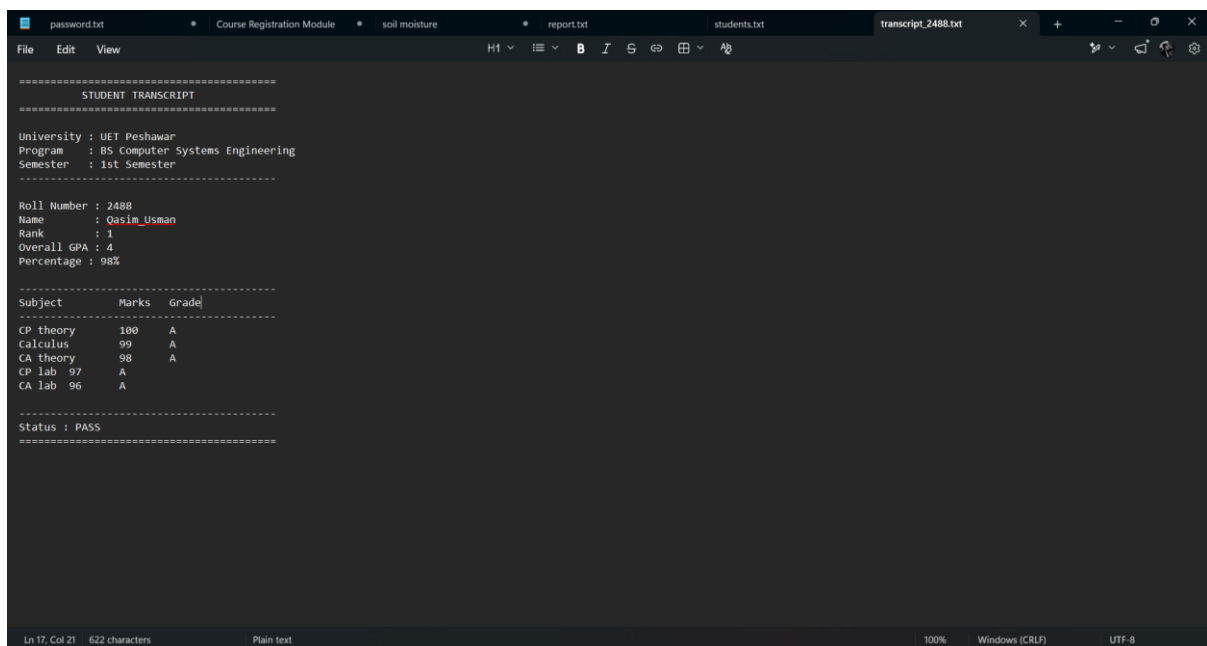
```
D:\UET Peshawar\DCSE\Projex x + v
Students loaded: 10 | Subjects: CP Theory, Calculus, CA Theory, CP Lab, CA Lab

===== MAIN MENU =====
1. Display All Students (Ranked)
2. Search Student
3. Add New Student
4. Update Student Marks
5. Delete Student
6. Generate Class Report
7. Generate Transcript
8. Subject-wise Statistics
9. Exit
Choice: 7
Enter Roll Number: 2488
Transcript generated successfully!
File Name : transcript_2488.txt
```

Figure 11.9: Console Output for Transcript Generation

11.10 Generated Student Transcript File

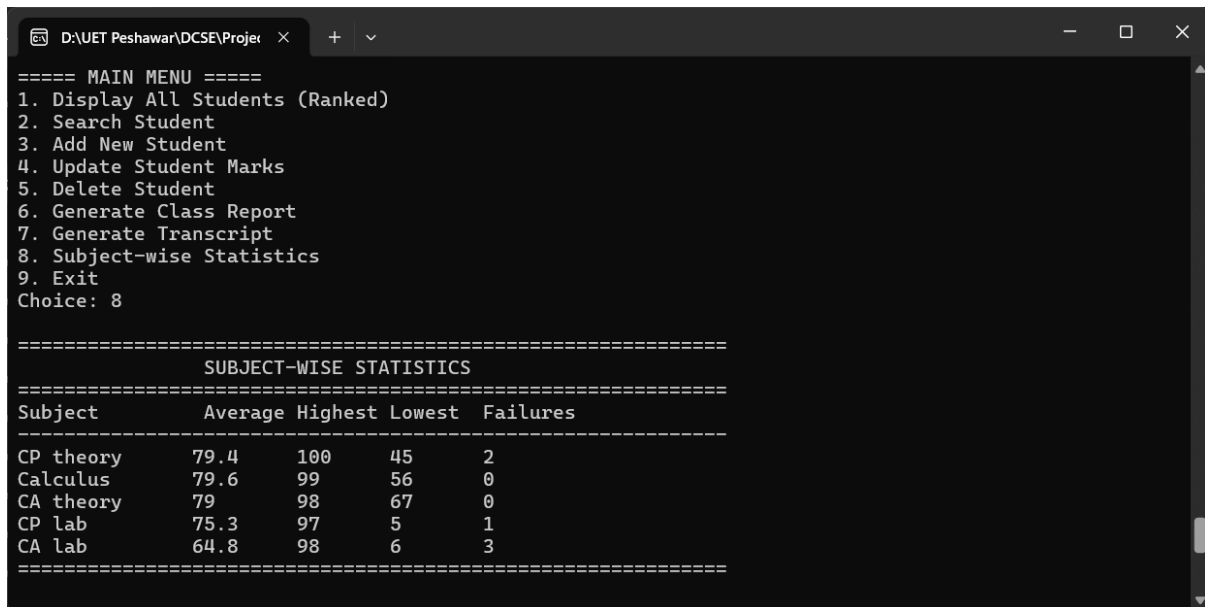
Contents of transcript_2488.txt.



```
=====
STUDENT TRANSCRIPT
=====
University : UET Peshawar
Program    : BS Computer Systems Engineering
Semester   : 1st Semester
-----
Roll Number : 2488
Name        : Qasim Usman
Rank        : 1
Overall GPA : 4
Percentage  : 98%
-----
Subject      Marks  Grade
-----
CP theory    100    A
Calculus     99     A
CA theory    98     A
CP lab       97     A
CA lab       96     A
-----
Status : PASS
=====
```

Figure 11.10: Generated Student Transcript

11.11 Subject-wise Statistics



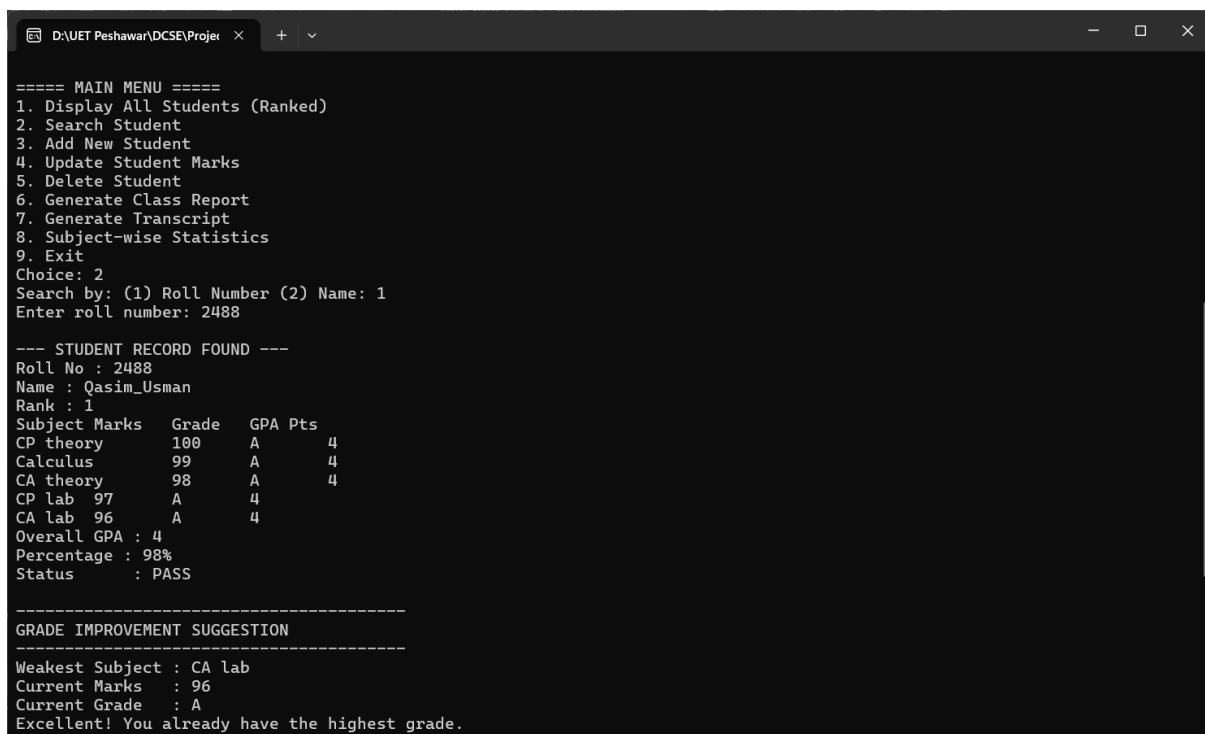
```
==== MAIN MENU ====
1. Display All Students (Ranked)
2. Search Student
3. Add New Student
4. Update Student Marks
5. Delete Student
6. Generate Class Report
7. Generate Transcript
8. Subject-wise Statistics
9. Exit
Choice: 8

=====
SUBJECT-WISE STATISTICS
=====
Subject      Average Highest Lowest Failures
-----
CP theory    79.4      100     45      2
Calculus     79.6       99     56      0
CA theory    79         98     67      0
CP lab       75.3       97      5      1
CA lab       64.8       98      6      3
=====
```

Figure 11.11: Subject-wise Statistics

11.12 Grade Improvement Suggestion

Displays the student's weakest subject, current grade, target grade, required marks for improvement, and a personalized academic recommendation.



```
==== MAIN MENU ====
1. Display All Students (Ranked)
2. Search Student
3. Add New Student
4. Update Student Marks
5. Delete Student
6. Generate Class Report
7. Generate Transcript
8. Subject-wise Statistics
9. Exit
Choice: 2
Search by: (1) Roll Number (2) Name: 1
Enter roll number: 2488

--- STUDENT RECORD FOUND ---
Roll No : 2488
Name : Qasim_Usman
Rank : 1
Subject Marks   Grade   GPA Pts
CP theory      100     A      4
Calculus       99      A      4
CA theory      98      A      4
CP lab 97      A      4
CA lab 96      A      4
Overall GPA : 4
Percentage : 98%
Status : PASS

-----
GRADE IMPROVEMENT SUGGESTION
-----
Weakest Subject : CA lab
Current Marks : 96
Current Grade : A
Excellent! You already have the highest grade.
```

Figure 11.12: Grade Improvement Suggestion

12. Assessment Rubric

Task	Marks	Assessment Criteria
Student File Design & Loading	/10	Correct file format, minimum of 10 student records, 5 subjects per student, and proper file parsing.
Struct Definition (Student & Subject)	/10	Correct field types, appropriate data organization, and effective use of nested structures or arrays.
GPA / Grade Calculation	/15	Accurate GPA calculation, correct grade boundaries, and proper calculation of subject-wise and overall GPA.
Add / Search / Update / Delete Students	/15	All four operations function correctly, including searching by both roll number and student name.
Ranked Report Generation (File Output)	/15	Students correctly sorted according to GPA, proper rank assignment, and successful generation of report.txt .
Failing / At-Risk Alert System	/10	Correct identification of failing or at-risk students with appropriate warning messages.
Input Validation & Error Handling	/10	Proper handling of missing files, duplicate roll numbers, invalid marks, and other user input errors.
Code Quality, Modularity & Comments	/5	Well-structured modular code, meaningful variable/function names, use of constants instead of magic numbers, and appropriate code comments.
Lab Report	/10	All required report sections included, file format documented, and program flow diagram provided.
Viva Voce	/10	Ability to explain the program, modify the code during viva, and answer design and implementation questions confidently.
Total	/100	Overall evaluation of project implementation, documentation, and viva performance.